

The Program in Action

When you run *sudoku.py*, the output will look like this:

```
Sudoku Puzzle, by Al Sweigart al@inventwithpython.com
--snip--
  A B C | D E F | G H I
1  . . . | . . . | . . .
2  . 7 9 | . 5 . | 1 8 .
3  8 . . | . . . | . . 7
   -----+-----+-----
4  . . 7 | 3 . 6 | 8 . .
5  4 5 . | 7 . 8 | . 9 6
6  . . 3 | 5 . 2 | 7 . .
   -----+-----+-----
7  7 . . | . . . | . . 5
8  . 1 6 | . 3 . | 4 2 .
9  . . . | . . . | . . .

Enter a move, or RESET, NEW, UNDO, ORIGINAL, or QUIT:
(For example, a move looks like "B4 9".)
--snip--
```

How It Works

Objects of the `SudokuGrid` class are the data structures that represent the Sudoku grid. You can call their methods to make modifications to, or retrieve information about, the grid. For example, the `makeMove()` method places a number on the grid, the `resetGrid()` method restores the grid to its original state, and `isSolved()` returns `True` if all the solution's numbers have been placed on the grid.

The main part of the program, starting on line 141, uses a `SudokuGrid` object and its methods for this game, but you could also copy and paste this class into other Sudoku programs you create to reuse its functionality.

```
1. """Sudoku Puzzle, by Al Sweigart al@inventwithpython.com
2. The classic 9x9 number placement puzzle.
3. More info at https://en.wikipedia.org/wiki/Sudoku
4. View this code at https://nostarch.com/big-book-small-python-projects
5. Tags: large, game, object-oriented, puzzle"""
6.
7. import copy, random, sys
8.
9. # This game requires a sudokupuzzle.txt file that contains the puzzles.
10. # Download it from https://inventwithpython.com/sudokupuzzles.txt
11. # Here's a sample of the content in this file:
12. # ..3.2.6..9..3.5..1..18.64....81.29..7.....8..67.82....26.95..8..2.3..9..5.1.3..
13. # 2...8.3...6..7..84.3.5..2.9...1.54.8.....4.27.6...3.1..7.4.72..4..6...4.1...3
14. # .....9.7...42.18....7.5.261..9.4....5.....4....5.7..992.1.8....34.59....5.7.....
15. # .3..5..4...8.1.5..46.....12.7.5.2.8....6.3....4.1.9.3.25.....98..1.2.6...8..6..2.
16.
```